What CogLean proves

Laws and theorems, module by module

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How to read it

- **Laws** are phenomena defined once, theory-neutrally, in the library (`CogLib.Phenomena`); a module *has* a law, *never* has it, *lacks* it (one counterexample), or *can hold* it (one case).
- **General** theorems bind variables: every net, problem, number or setting, under stated hypotheses. **Instances** are single cases computed exactly.
- Generated from Lean (`tools/gen_catalog.py` → `notes/CATALOG.md`); every theorem uses only the standard axioms; no `sorry`. **Given**: the laws a theorem assumes (32 are conditional on laws).

	Theorems	General	Instances	About laws	Headline
CogLib (the library)	198	193	5	24	78
EPAM	31	14	17	7	18
Strategy choice	20	15	5	0	11
Multi-digit (with problem solver)	166	139	27	1	67
Theory matrix	61	46	15	31	30
Experiments	45	25	20	0	13
Concepts	17	12	5	5	8
Semantic memory	13	6	7	6	7
View application	14	5	9	0	8
All	565	455	110	74	240

The library: laws, and how they relate

Laws defined in CogLib

- Learning: non-interference, criterion, generalization, serial position
- Responding: oscillation, retroactive interference
- Retrieval: certainty, stability, order-freedom, recency, stored-only, determinacy
- Verification: distance, typicality, reversal
- Combination: conjunction effect, over-extension, dissociation

Proved about the laws themselves

- Non-interference $\Rightarrow$ learning to criterion
- Order-freedom $\Rightarrow$ never recency
- Determinacy + certainty + stability $\Rightarrow$ confident errors
- Reversal $\Rightarrow$ no distance effect
- Conjunction effect $\Rightarrow$ over-extension
- Membership monotone in typicality $\Rightarrow$ never dissociation
- Attention nearer an anchor first $\Rightarrow$ serial position effect *and* isolation effect

The library: general results about its mechanisms

- **Discrimination nets**: recognition depends only on features tested; learning never disturbs another item; nets have non-interference, generalization, oscillation, retroactive interference.
- **Sampled retrieval**: used with probability $1 - (1 - R)^n$; error share independent of search length.
- **Semantic nets**: inheritance with override and accumulation; distance is the least level.
- **View application**: the view is kept; the view wins on matched frames; unmatched frames survive; instances are re-parented.
- **Spreading activation**: sources sum; priming never lowers activation or a candidate's share.
- **Problem solving**: means-ends analysis is sound for any space; how it chains and fails; runs traced as observables.

EPAM, and memory families against retrieval laws

EPAM has, for every net and list: non-interference, learning to criterion, stimulus generalization, oscillation, retroactive inhibition, serial position.

On List A (instance): stimulus generalization.

	Cert.	Stab.	Order-free	Recency	Stored	Determ.
Discrimination	**has**	**has**	lacks	lacks	**has**	**has**
Activation	**never**	**never**	**never**	**has**	**has**	**never**
Strength	lacks	lacks	**has**	**never**	**has**	lacks
Global matching	lacks	lacks	**has**	**never**	lacks	**has**

Strategy choice (Siegler & Shrager 1984)

General, for any associations, criterion and search length:

- retrieval is used with probability $1 - (1 - R)^n$;
- the retrieval error share is the wrong answers' share above the criterion, whatever the search length;
- retrieval does not depend on the backup strategies;
- the error share on counting trials involves neither the associations nor the criterion;
- learning from a correct answer, or moving strength to it, never raises retrieval errors.

Instances: predicted overt-strategy order across six problems; peaking can *raise* overt use (a limit the chapter does not state).

Multi-digit addition: recognition and retrieval

General, with any memory (EPAM's net or any retriever):

- **confident errors**: a confused problem errs on every retrieval, with no extra backup use;
- a confused problem gets its neighbour's answer; same-sum and commuted confusions are harmless;
- strategy nesting; practiced problems stay practiced;
- near misses within 200, or 20; confusions keep zeros, thresholds and column sums that are tested;
- discrimination removes a confusion, and only that; then the columns;
- the integrated application (solver as backup, DOA facts): whole-problem errors are untouched by the solver, and a solver error is a single-digit error in its column.

Instances: the worked cases (EPAM 45/46; ACT-R graded, and lacking *agrees-on*); with every answer computed, $150 + 100$ is correct $\approx .014$ (EPAM), $\approx .238$ (ACT-R).

Multi-digit addition: the column backup as problem solving

General, for every problem:

- **the bridge**: the problem solver answers exactly as the column backup, for every single-digit process;
- place value: carrying adds no error; a column error is a single-digit error moved to its place;
- without the carry rule: exact when no carry is needed; under strict critics, the no-carry bug;
- GPS works units first and answers exactly; without the carry operator it gives up on exactly the carry problems;
- **the same deficit, two regimes**: a confident bug under repair, no answer under GPS.

Instance: four repairs, four bugs on $15 + 17$.

Experiments written against the memory interface

General:

- recency and order-freedom are inconsistent wherever a valid design exists;
- order-freedom gives no order effect, recency a positive one; under forced choice, order-freedom reports 0 however biased the guessing;
- a flat serial-position curve, from certainty and stability alone;
- confident misnaming, derived from EPAM's matrix cells.

Instances: ACT-R's rising curve; MINERVA returns the never-studied prototype; ACT-R says "don't know" more to strangers.

Concepts and semantic memory

Concepts (selective modification + threshold), general:

- conjunction effect; its reverse;
- the attribute, not the value;
- over-extension (yes/no; graded);
- every even number is more typical of *even multiple of 3*.

Instances: the paper's figures; 14 at .80 vs .06.

Semantic memory:

- Hierarchical (C&Q): distance **has**, typicality **never**, reversal **never**; parallel lines (all general).
- Feature comparison (SSR): more similar is never slower (general); typicality and reversal can hold, distance lacks (instances).

View application

General:

- (library) the view is kept; the view wins on matched frames; unmatched frames survive; instances re-parented;
- after the ARITHMETIC view, membership in *even multiple of 3* is exactly divisibility by 6, for every number;
- no over-extension after the view, at any number; typicality unchanged, for every number.

Instances:

- Table 4 reproduced (probability 1/2);
- before the view, over-extension at 14; after, dissociation (14 vs 9);
- view selection: "+" read as adding 2/3 (naive), as appending a step 3/5 (with a memory).

Derived, or close to the definitions? (the audit)

Compositional	confident errors (recognition + associations); after the view (view application + categorization); the two regimes diverge
Derived	no-carry bug; GPS order; isolation effect; DOA's counting error share; EPAM's oscillation
Close	serial position (now from a rule); conjunction effect; DOA's closed forms
Definitional	hierarchical network's costs; algorithm properties (soundness, merges)

A judgment, not a check (`notes/EXPLANATORY_AUDIT.md`). Its weakest results are its targets.

Where the results are weakest

- **Single counterexamples**: eight "lacks" cells in the memory matrix; the feature model's verdicts; ACT-R's in the multi-digit cases.
- **Examples only**: view selection's readings; Table 4; the four bugs on $15 + 17$.
- **Chosen settings**: worked cases' strengths and bounds; priors not yet from experience.
- **Not yet formalized**: Soar's chunking and other impasses; Collins & Loftus's decision process; subtraction.

The catalog is regenerated from Lean, so these lists shrink as general proofs replace examples.